

The background of the slide is a dark, semi-transparent financial chart. It features a candlestick chart with green and red bars, overlaid with several moving average lines in blue, green, and yellow. The chart is set against a dark grid.

INVESTOR SENTIMENT

AND THE PRICING OF

BEHAVIORAL BETA

IN

U.S. EQUITY MARKETS



What is the Investor Sentiment & the Pricing of Behavioral Beta?

- Traditional finance models explain stock returns mainly through market risk. These models assume that investors are rational and that financial markets are efficient. However, in real markets, investors are often influenced by emotions such as optimism, fear, and overconfidence, which may affect their investment decisions.
- This study examines whether investor sentiment affects stock returns in the U.S. equity market. In particular, it focuses on behavioural beta, which measures how sensitive asset returns are to changes in investor sentiment.
- Using monthly data on market returns, ETF returns, and a sentiment index, this research investigates whether sentiment-related risk is priced in financial markets. The study also evaluates whether incorporating behavioural factors improves traditional asset pricing models.
- The aim of this research is to assess the extent to which behavioural factors contribute to asset pricing and to determine whether traditional financial models remain effective in explaining returns.





Research

Objective & Scope



Objectives:

- To examine whether investor sentiment affects ETF returns in the U.S. equity market.
- To estimate behavioural beta as a measure of sensitivity to investor sentiment.
- To analyse whether sentiment-related risk is priced in asset returns.
- To evaluate whether traditional asset pricing models improve when sentiment is included.

Scope of the Study:

- Market: U.S. Equity Market
- Data: Monthly returns of SPX, selected ETFs, and investor sentiment index
- Period: Approximately 23–24 years (monthly data)
- Method: Regression and asset pricing analysis
- Focus: Investor sentiment and behavioural factors



Data Sources

Component	Sources	Frequency
Sector ETF Returns (XLE, XLF, XLP, XLK, XLU)	Market data (Yahoo Finance)	Monthly
S&P 500 Return (SPX)	Market data (Yahoo Finance)	Monthly
Sentiment Index	Baker–Wurgler method	Monthly
Fama–French Factors	Kenneth French data library	Monthly





Theoretical Framework



Framework:

Investor Sentiment → Behavioural Beta → Asset Returns
Market Risk → Asset Returns

- Traditional finance theory, particularly the Capital Asset Pricing Model (CAPM), explains stock returns based on market risk, measured by beta. According to these models, investors are rational and markets efficiently incorporate all available information into asset prices.
- Behavioural finance challenges these assumptions by recognizing that investors are influenced by emotions, psychological biases, and social factors. These behavioural influences may lead to temporary mispricing in financial markets.
- Investor sentiment represents the overall level of optimism or pessimism among market participants. Changes in sentiment may influence trading behaviour and asset prices.
- Assets differ in their sensitivity to investor sentiment, which is referred to as behavioural beta. A higher behavioural beta indicates greater exposure to sentiment-driven risk.
- This study proposes that investor sentiment may influence asset returns both directly and through behavioural beta. However, the extent to which this behavioural risk is reflected in asset prices remains an empirical question, which this research aims to examine.



Statistical Tools



Tool
OLS Regression
t-test / p-values
Robust SE (HC3)
R^2
Durbin-Watson
Jarque-Bera
Mean test (Spread)
Interaction term
Pooled regression
Descriptive statistics

Purpose in Thesis
Estimate relationships
Test statistical significance
Correct heteroskedasticity
Model fit evaluation
Autocorrelation check
Normality test
Pricing test
Regime test
Joint pricing test
Data understanding





Methodology

Construction of Excess Returns

Components Used:

- ETF returns
- Market returns (SPX)
- Risk-free rate

Procedure:

- Subtract risk-free rate from ETF returns
- Subtract risk-free rate from market returns
- Use excess returns for all analyses

Purpose:

- Measure performance beyond risk-free investment

Equation:

$$R(\text{excess})_{i,t} = R_{i,t} - R_{f,t}$$

$$R(\text{excess})_{m,t} = R_{m,t} - R_{f,t}$$



CAPM Analysis

Market Risk Estimation (CAPM)

Components:

- ETF excess returns (dependent variable)
- SPX excess returns (market factor)

Method:

- Regress ETF excess returns on market returns
- Estimate market beta and model fit

Objective:

- Measure each ETF's sensitivity to market movements
- Establish a benchmark pricing model

Equation:

$$R_{i,t}^{\text{excess}} = \alpha_i + \beta_{m,i} R_{m,t}^{\text{excess}} + \varepsilon_{i,t}$$

Hypothesis (H1)

ETFs are significantly related to market movements.

Results

- All ETFs show positive and significant market betas.
- Betas range from 0.50 (XLU) to 1.28 (XLK).
- R^2 values range from 25% to 74%.

Interpretation

- US sector ETFs are strongly influenced by market risk.
- Technology (XLK) and Financials (XLF) are most sensitive.
- Utilities (XLU) is least sensitive.

Conclusion

Market risk is the primary driver of ETF returns.



CAPM Analysis **Results**

ETF	Market Beta	p-value	R-squared
XLE	1.059716	5.302877e-23	0.398879
XLF	1.180328	1.893576e-93	0.702324
XLP	0.547586	8.885279e-42	0.454607
XLK	1.284772	1.009251e-98	0.742609
XLU	0.501417	1.410729e-16	0.251956





Behavioural Beta Estimation

Measuring Sensitivity to Investor Sentiment

Components:

- ETF excess returns
- Market excess returns
- Orthogonalised sentiment index

Method:

- Include sentiment together with market returns
- Estimate behavioural beta for each ETF

Objective:

- Identify how ETF returns react to changes in sentiment
- Measure behavioural risk exposure

Equation:

$$R_{i,t}^{\text{excess}} = \alpha_i + \beta_{m,i} R_{m,t}^{\text{excess}} + \beta_{s,i} \text{Sentiment}_t + \varepsilon_{i,t}$$

Hypothesis (H2)

Investor sentiment significantly affects ETF returns.

Results

- Only XLE shows significant behavioural beta ($p < 0.05$).
- Other ETFs show insignificant sentiment coefficients.
- Behavioural betas are generally small.

Interpretation

- Energy sector is more influenced by investor sentiment.
- Most ETFs are dominated by fundamentals and market forces.

Conclusion

Limited evidence of sentiment impact on ETF pricing.





Behavioural Beta Estimation **Results**

ETF	Alpha	Market Beta	Behavioural Beta	Sentiment p-value	R-squared
XLE	0.001189	1.090227	0.012547	0.034946	0.411909
XLF	-0.000646	1.192162	0.004867	0.157598	0.705106
XLP	0.002750	0.552486	0.002015	0.522289	0.456041
XLK	0.002507	1.269994	-0.006077	0.228401	0.746481
XLU	0.003236	0.503519	0.000864	0.801695	0.252130



Predictive Regression Analysis

Sentiment and Future Returns

Components:

- Current sentiment values
- Next-month ETF excess returns

Method:

- Shift ETF returns forward by one month
- Regress future returns on current sentiment

Objective:

- Test whether sentiment predicts future market performance
- Examine market inefficiency

Equation:

$$R_{i,t+1}^{\text{excess}} = \alpha_i + \gamma_i \text{Sentiment}_t + \varepsilon_{i,t+1}$$

Hypothesis (H3)

Investor sentiment predicts future ETF returns.

Results

- No ETF shows strong predictive power.
- XLK is weakly significant (10% level).
- Most p-values > 0.25.

Interpretation

- Sentiment does not consistently forecast returns.
- Market is mostly efficient at monthly frequency.

Conclusion

Investor sentiment has weak predictive ability.



Predictive Regression Analysis **Results**

ETF	Predict Coeff	p-value
XLE	0.003148	0.629537
XLF	-0.005702	0.275771
XLP	-0.003492	0.290200
XLK	-0.016139	0.079060
XLU	-0.004006	0.299015





Portfolio and Spread Analysis

High–Low Behavioural Beta Test

Components:

- Behavioural beta estimates
- ETF excess returns

Method:

- Rank ETFs by behavioural beta
- Identify highest and lowest sensitivity ETFs
- Create return spread between them
- Test average spread return

Objective:

- Examine whether sentiment risk is rewarded
- Test pricing of behavioural beta

Equation:

$$\text{Spread}_t = R^{\text{excess}}_{\text{High},t} - R^{\text{excess}}_{\text{Low},t}$$

$$\text{Spread}_t = \alpha + \varepsilon_t$$

Hypothesis (H4)

ETFs with high behavioural beta earn higher returns.

Results

- Spread alpha = 0.0009
- Not statistically significant ($p = 0.84$).

Interpretation

- No return premium for sentiment risk.
- Investors are not rewarded for sentiment exposure.

Conclusion

Behavioural beta is not priced in ETF returns.





Portfolio and Spread Analysis **Results**

```
=== High-Low Spread Test ===  
                                OLS Regression Results  
=====
```

Dep. Variable:	Spread	R-squared:	0.000
Model:	OLS	Adj. R-squared:	0.000
Method:	Least Squares	F-statistic:	nan
Date:	Sun, 08 Feb 2026	Prob (F-statistic):	nan
Time:	10:48:43	Log-Likelihood:	331.71
No. Observations:	288	AIC:	-661.4
Df Residuals:	287	BIC:	-657.8
Df Model:	0		
Covariance Type:	HC3		

```
=====
```

	coef	std err	z	P> z	[0.025	0.975]
const	0.0009	0.005	0.197	0.844	-0.008	0.010

```
=====
```

Omnibus:	13.963	Durbin-Watson:	2.037
Prob(Omnibus):	0.001	Jarque-Bera (JB):	27.820
Skew:	0.213	Prob(JB):	9.10e-07
Kurtosis:	4.462	Cond. No.	1.00

```
=====
```

Notes:
[1] Standard Errors are heteroscedasticity robust (HC3)





Regime and Interaction Analysis

Market Condition Effects

Components:

- Investor sentiment
- Market volatility indicator
- Interaction variable

Method:

- Identify high-volatility periods
- Interact sentiment with volatility
- Estimate joint effects

Objective:

- Test whether sentiment effects are stronger during unstable markets
- Capture behavioural effects in crisis periods

Equation:

$$R_{i,t}^{\text{excess}} = \alpha_i + \beta_{m,i} R_{m,t}^{\text{excess}} + \beta_{s,i} \text{Sentiment}_t + \beta_{\text{int},i} (\text{Sentiment}_t \times \text{HighVol}_t) + \varepsilon_{i,t}$$

Hypothesis (H5)

Sentiment effects increase during high volatility periods.

Results

- Interaction term insignificant for all ETFs.
- No strong regime dependence.

Interpretation

- Market stress does not amplify sentiment impact.
- Professional trading limits behavioural biases.

Conclusion

Sentiment influence is stable across market conditions.





Regime and Interaction Analysis **Results**

ETF: XLE

OLS Regression Results

```
=====
Dep. Variable:          XLE_excess    R-squared:                0.414
Model:                  OLS           Adj. R-squared:           0.407
Method:                 Least Squares  F-statistic:              33.75
Date:                  Sun, 08 Feb 2026  Prob (F-statistic):      1.10e-18
Time:                  10:48:43        Log-Likelihood:           415.06
No. Observations:      288            AIC:                     -822.1
Df Residuals:          284            BIC:                     -807.5
Df Model:               3
Covariance Type:       HC3
=====
```

	coef	std err	z	P> z	[0.025	0.975]
const	0.0012	0.003	0.363	0.717	-0.005	0.008
SPX_excess	1.0977	0.110	9.992	0.000	0.882	1.313
SENT_ORTH	0.0083	0.007	1.249	0.212	-0.005	0.021
Interaction	0.0088	0.012	0.707	0.480	-0.016	0.033

```
=====
Omnibus:                22.881    Durbin-Watson:            2.120
Prob(Omnibus):           0.000    Jarque-Bera (JB):         55.132
Skew:                   0.349     Prob(JB):                 1.07e-12
Kurtosis:                5.026    Cond. No.                  23.6
=====
```

Notes:

[1] Standard Errors are heteroscedasticity robust (HC3)





Regime and Interaction Analysis **Results**

```
ETF: XLF

                                OLS Regression Results
=====
Dep. Variable:                  XLF_excess    R-squared:                0.707
Model:                          OLS          Adj. R-squared:           0.704
Method:                        Least Squares  F-statistic:             147.2
Date:                          Sun, 08 Feb 2026  Prob (F-statistic):    1.51e-57
Time:                          10:48:43      Log-Likelihood:          565.56
No. Observations:              288          AIC:                     -1123.
Df Residuals:                  284          BIC:                     -1108.
Df Model:                      3
Covariance Type:               HC3
=====
               coef    std err          z      P>|z|      [0.025    0.975]
-----
const          -0.0006     0.002    -0.301     0.764    -0.005     0.004
SPX_excess      1.1993     0.060    19.943     0.000     1.081     1.317
SENT_ORTH       0.0008     0.004     0.179     0.858    -0.008     0.009
Interaction     0.0084     0.007     1.222     0.222    -0.005     0.022
=====
Omnibus:                24.034    Durbin-Watson:           1.957
Prob(Omnibus):           0.000    Jarque-Bera (JB):        95.191
Skew:                   -0.047    Prob(JB):                2.14e-21
Kurtosis:                5.815    Cond. No.                 23.6
=====

Notes:
[1] Standard Errors are heteroscedasticity robust (HC3)
```





Regime and Interaction Analysis **Results**

ETF: XLP

OLS Regression Results

Dep. Variable:	XLP_excess	R-squared:	0.457
Model:	OLS	Adj. R-squared:	0.451
Method:	Least Squares	F-statistic:	66.35
Date:	Sun, 08 Feb 2026	Prob (F-statistic):	1.52e-32
Time:	10:48:43	Log-Likelihood:	635.13
No. Observations:	288	AIC:	-1262.
Df Residuals:	284	BIC:	-1248.
Df Model:	3		
Covariance Type:	HC3		

	coef	std err	z	P> z	[0.025	0.975]
const	0.0028	0.002	1.794	0.073	-0.000	0.006
SPX_excess	0.5553	0.040	13.988	0.000	0.477	0.633
SENT_ORTH	0.0004	0.005	0.091	0.927	-0.009	0.009
Interaction	0.0033	0.007	0.502	0.616	-0.010	0.016

Omnibus:	12.995	Durbin-Watson:	1.990
Prob(Omnibus):	0.002	Jarque-Bera (JB):	27.155
Skew:	-0.153	Prob(JB):	1.27e-06
Kurtosis:	4.473	Cond. No.	23.6

Notes:

[1] Standard Errors are heteroscedasticity robust (HC3)





Regime and Interaction Analysis **Results**

ETF: XLK

OLS Regression Results

```
=====
Dep. Variable:          XLK_excess    R-squared:                0.750
Model:                  OLS           Adj. R-squared:           0.747
Method:                 Least Squares F-statistic:              162.4
Date:                   Sun, 08 Feb 2026 Prob (F-statistic):      2.59e-61
Time:                   10:48:43      Log-Likelihood:          571.78
No. Observations:       288           AIC:                    -1136.
Df Residuals:           284           BIC:                    -1121.
Df Model:                3
Covariance Type:        HC3
=====
```

	coef	std err	z	P> z	[0.025	0.975]
const	0.0025	0.002	1.350	0.177	-0.001	0.006
SPX_excess	1.2607	0.060	21.035	0.000	1.143	1.378
SENT_ORTH	-0.0008	0.007	-0.101	0.919	-0.015	0.014
Interaction	-0.0110	0.010	-1.058	0.290	-0.031	0.009

```
=====
Omnibus:                40.345      Durbin-Watson:           2.213
Prob(Omnibus):          0.000      Jarque-Bera (JB):        144.332
Skew:                   0.528      Prob(JB):                4.56e-32
Kurtosis:               6.304      Cond. No.                23.6
=====
```

Notes:

[1] Standard Errors are heteroscedasticity robust (HC3)





Regime and Interaction Analysis **Results**

ETF: XLU

OLS Regression Results

```
=====
Dep. Variable:          XLU_excess    R-squared:                0.252
Model:                  OLS          Adj. R-squared:           0.244
Method:                 Least Squares F-statistic:              22.73
Date:                  Sun, 08 Feb 2026 Prob (F-statistic):      3.21e-13
Time:                  10:48:43      Log-Likelihood:          529.40
No. Observations:      288          AIC:                    -1051.
Df Residuals:          284          BIC:                    -1036.
Df Model:               3
Covariance Type:       HC3
=====
```

	coef	std err	z	P> z	[0.025	0.975]
const	0.0032	0.002	1.364	0.172	-0.001	0.008
SPX_excess	0.5037	0.062	8.168	0.000	0.383	0.625
SENT_ORTH	0.0008	0.005	0.151	0.880	-0.009	0.011
Interaction	0.0002	0.007	0.027	0.979	-0.013	0.014

```
=====
Omnibus:                2.631    Durbin-Watson:           2.037
Prob(Omnibus):           0.268    Jarque-Bera (JB):        2.356
Skew:                   -0.157    Prob(JB):                0.308
Kurtosis:                3.312    Cond. No.                23.6
=====
```

Notes:

[1] Standard Errors are heteroscedasticity robust (HC3)



Multi-Factor Robustness Analysis **Fama-French Five- Factor Model**

Components:

- Market factor
- Size factor (SMB)
- Value factor (HML)
- Profitability factor (RMW)
- Investment factor (CMA)
- Sentiment index

Method:

- Include all risk factors with sentiment
- Re-estimate behavioural effects

Objective:

- Control for traditional risk factors
- Confirm robustness of sentiment results

Equation:

$$R^{\text{excess}}_{i,t} = \alpha_i + \beta_m R^{\text{excess}}_{m,t} + \beta_{\text{smb}} \text{SMB}_t + \beta_{\text{hml}} \text{HML}_t + \beta_{\text{rmw}} \text{RMW}_t + \beta_{\text{cma}} \text{CMA}_t + \beta_s \text{Sentiment}_t + \varepsilon_{i,t}$$

Hypothesis (H6)

Sentiment remains significant after controlling for risk factors.

Results

- All sentiment coefficients insignificant.
- Slight effect for XLE (10% level).
- R^2 remains high.

Interpretation

- Traditional risk factors explain most returns.
- Sentiment adds little additional information.

Conclusion

Sentiment effects are not robust.



Multi-Factor Robustness Analysis **Results**

ETF	Sentiment Coefficient	Sentiment p-value	R-squared
XLE	0.010887	0.084075	0.426562
XLF	0.003369	0.372912	0.711335
XLP	0.001133	0.717056	0.468551
XLK	-0.004739	0.381138	0.748713
XLU	0.001726	0.655186	0.256902



Economic Significance and Panel Analysis

Economic Impact and Pooled Regression

Components:

- Standard deviation of sentiment
- Stacked ETF data
- Market and sentiment variables

Method:

- Measure volatility of sentiment
- Combine all ETFs into pooled dataset
- Estimate average behavioural effect

Objective:

- Evaluate economic importance of sentiment
- Improve statistical power

Equation:

$$R_{i,t}^{\text{excess}} = \alpha + \beta_m R_{m,t}^{\text{excess}} + \beta_s \text{Sentiment}_t + \varepsilon_{i,t}$$

Hypothesis (H7)

Sentiment has a significant effect at the market level.

Results

- Market factor highly significant.
- Sentiment insignificant ($p = 0.17$).
- $R^2 = 48\%$.

Interpretation

- Overall ETF market driven by market risk.
- Sentiment plays minor role.

Conclusion

No strong aggregate sentiment pricing effect.



Economic Significance and Panel Analysis **Results**

Standard Deviation of Sentiment:
0.6900498484603167

OLS Regression Results

```
=====
Dep. Variable:          excess      R-squared:          0.483
Model:                  OLS        Adj. R-squared:       0.482
Method:                 Least Squares  F-statistic:        363.8
Date:                  Sun, 08 Feb 2026  Prob (F-statistic):  1.42e-128
Time:                  10:48:43      Log-Likelihood:     2514.0
No. Observations:      1440         AIC:                -5022.
Df Residuals:          1437         BIC:                -5006.
Df Model:               2
Covariance Type:       HC3
=====
```

	coef	std err	z	P> z	[0.025	0.975]
const	0.0018	0.001	1.648	0.099	-0.000	0.004
SPX_excess	0.9217	0.034	26.921	0.000	0.855	0.989
SENT_ORTH	0.0028	0.002	1.371	0.170	-0.001	0.007

```
=====
Omnibus:                146.063      Durbin-Watson:        2.055
Prob(Omnibus):           0.000      Jarque-Bera (JB):     880.666
Skew:                   0.242      Prob(JB):             5.83e-192
Kurtosis:               6.800      Cond. No.             23.2
=====
```

Notes:

[1] Standard Errors are heteroscedasticity robust (HC3)



Does Market Beta Influences Behavioural Beta ?



Hypothesis (H7)

Investor sentiment has a significant independent effect on ETF returns, even without controlling for market risk.

Results

- Sentiment coefficients (Behavioural Beta) are:
- XLE: 0.00145 ($p = 0.83$) → Not significant
- XLF: -0.00727 ($p = 0.16$) → Not significant
- XLP: -0.00361 ($p = 0.31$) → Not significant
- XLK: -0.01900 ($p = 0.038$) → Significant
- XLU: -0.00426 ($p = 0.27$) → Not significant
- R^2 values are very low (below 4% for all ETFs).

Interpretation

The behavioural beta-only regression indicates that investor sentiment does not significantly explain ETF excess returns in the absence of market controls. With the exception of XLK, sentiment coefficients are statistically insignificant and R^2 values remain very low. This suggests that investor sentiment does not independently price sector ETFs, and its effect is largely conditional on market risk factors.

Conclusion

- Investor sentiment cannot independently price ETF returns without considering market risk.
- Market factors are essential in explaining returns.
- Behavioural beta becomes meaningful only when combined with market variables.
- Technology stocks appear more sensitive to sentiment than other sectors.
- “Overall, the evidence suggests that while investor sentiment may influence certain sectors, behavioural beta is not systematically priced in U.S. equity sector ETFs.”

=== Behavioural Beta ONLY (No Market) ===

	ETF	Alpha	Behavioural_Beta	Sentiment_pvalue	R_squared
0	XLE	0.007267	0.001451	0.828665	0.000179
1	XLF	0.006001	-0.007267	0.157196	0.006360
2	XLP	0.005830	-0.003608	0.307011	0.004716
3	XLK	0.009588	-0.019003	0.037780	0.038816
4	XLU	0.006044	-0.004260	0.266389	0.004346



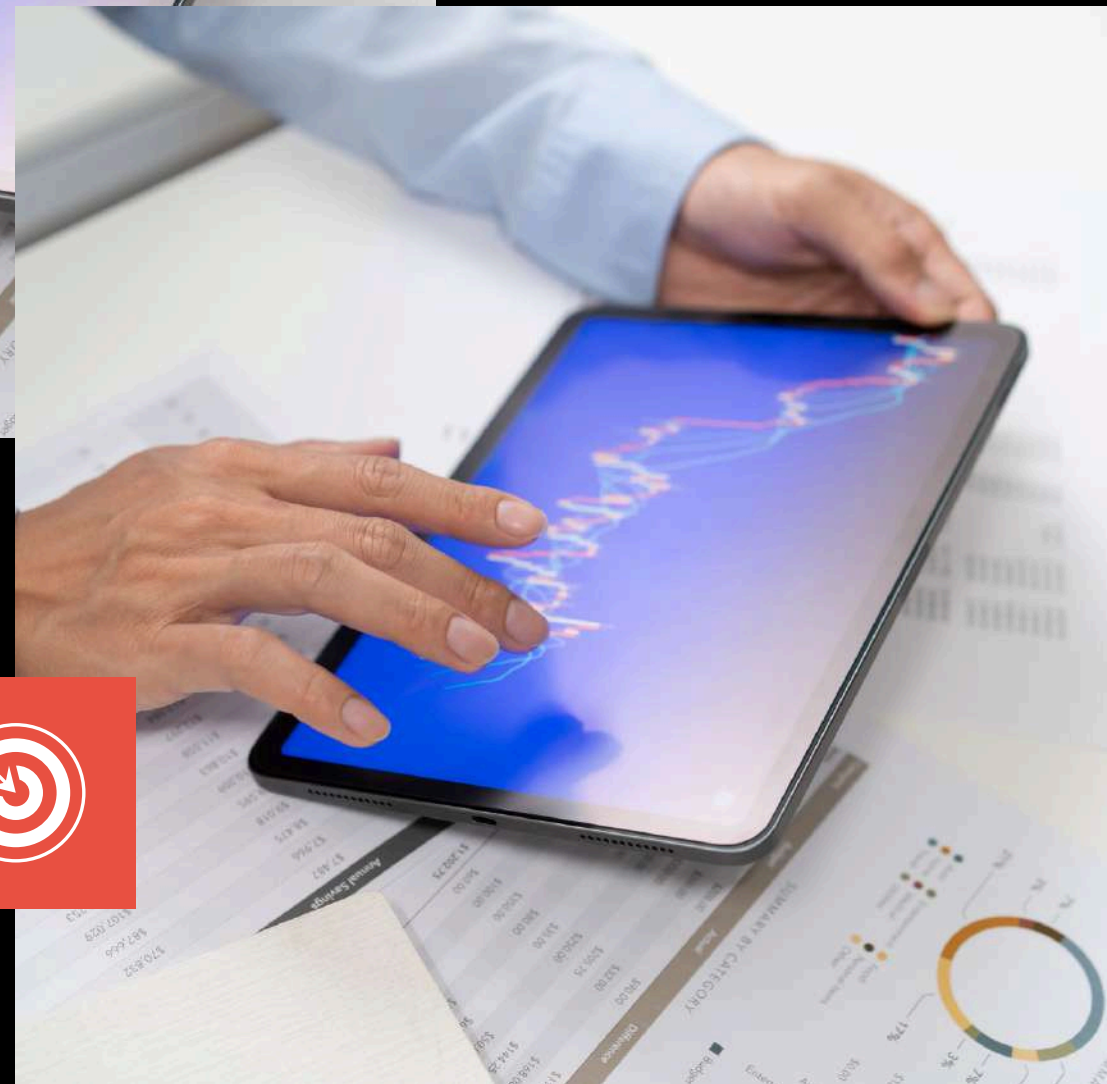
Conclusion

Key Findings

- Market beta strongly explains ETF returns.
- Behavioural beta is mostly insignificant.
- No consistent pricing of sentiment risk.
- Weak evidence of predictability.

Implications

- US ETF market is largely efficient.
- Institutional investors reduce sentiment distortions.
- Traditional asset pricing models remain valid.





**THANK
YOU**

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